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To: Ms. Alysoun Taylor-Hall

From: Section 07, Group 3, Ustav Acharya, Daniel Evans, Erica Gilliam, Reese Harmeyer, Carter Rockenbaugh

Subject: Improving Floating Solar Farms in the Ocean

Date: April 4, 2026

Purpose

We propose to improve the efficiency and durability of floating solar farms in the ocean to make them a more viable power solution. *Excellent*

Summary

Solar power is a very well-known source of renewable energy. Unfortunately, solar panels often take up land that is needed for other things such as agriculture. But with the use of floating solar farms, that scarce land can be saved for more important things. Floating solar farms (FSFs) are emerging to be a viable option for large scale solar power production [1]. By putting solar panels into water, the solar panels efficiency is boosted due to the water keeping the panels cool.

Floating solar farms have only had major success on still water such as lakes or ponds and even fish farms. These farms have had trouble being placed successfully in the ocean. The idea becomes more complex due to the salt water and stronger waves causing faster wear and tear on the panels and floatation platforms that they sit on. The salt water also dirties the panels themselves and if the complete setup is not maintained properly, it will not be able to generate enough power [2].

Avoid overpromising

With \$250,000 our team plans to solve these issues to make floating farms more available for ocean use. Our team will do research on the best locations to install solar farms in the ocean. This will help with the placement of our anchoring and mooring system that will help to keep the floating panels in place and weaken the strength of incoming waves on the platforms. The team will also do research on a cleaning method to keep the panels clear of contaminants to ensure maximum power is being collected. This research will reduce long term cost of installing floating farms in the ocean as it will keep the farm in working order for longer.

Qualifications

Exact scope? Prototype? testing?

Carter R. is a mechanical engineer who has experience in analyzing how systems perform technically and economically. Carter will be responsible for finding a suitable location and weighing the costs of making our solar farm better. Daniel E. is also a mechanical engineer with knowledge of mechanical properties of materials and structural designs. He will be responsible for selecting suitable material and floatation systems for the solar panels. Utsav Acharya is a Computer Science student with experience in software development using Java, Python, JavaScript, and PHP. He will be responsible for researching and developing the software-based dirty panel monitoring and detection system for the floating solar farm. Reese H. is a computer engineer with experience installing and working on electrical systems. Reese will be responsible for the development of the cleaning system that will work with a detections system for the

good

Good work. Work on clarifying exact scope

panels. Erica G. is an industrial engineer with experience in modeling and data analysis. Erica will be responsible for analyzing the materials to find the most suitable for this application.

References

- [1] Sree, D. K. K., Law, A. W.-K., Pang, D. S. C., Tan, S. T., Wang, C. L., Kew, J. H., Seow, W. K., & Lim, V. H. (2022). Fluid-structural analysis of modular floating solar farms under wave motion. *Solar Energy*, 233, 161–181. <https://doi.org/10.1016/j.solener.2022.01.017>
- [2] Thangarasu, S. K., Gobinath, V. K., Shanmugam, A., Kumar, S. M., Shrinidhi, K. M., & Sanjay, S. (2024). Innovative Arduino-based solar panel cleaning system for enhanced energy efficiency. *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kamand, India*, 1–5. <https://doi.org/10.1109/ICCCNT61001.2024.10725904>

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Annotated Bibliography

Bru, J., Seland, T. S., Dai, J., & Jiang, Z. (2025). Life cycle cost analysis of an offshore floating

photovoltaic concept in the North Sea. *Renewable Energy*, 249, Article 122981.

<https://doi.org/10.1016/j.renene.2025.122981>

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This journal article explains how much it costs to maintain offshore floating solar panels and how long they last before they need to be replaced. We will use this information to help design solar farms that are more cost efficient. This source was submitted by Carter Rockenbaugh. 1

Hegarty, A., Westbrook, G., Glynn, D., Murray, D., Omerdic, E., & Toal, D. (2019, April 15–18).

A low-cost remote solar energy monitoring system for a buoyed IoT ocean observation platform [Conference session]. 2019 IEEE World Forum on Internet of

Things (WF-IoT), Limerick, Ireland. <https://doi.org/10.1109/WF-IoT.2019.8767311>

Conference
This paper discusses a solar monitoring system designed for ocean-based platforms. It shows how solar systems can be monitored remotely in harsh marine environments. ^{we} will use this source to directly connect monitoring systems to floating solar farms, showing how the system works in ocean conditions. This source was submitted by Utsav Acharya. 2

Huang, L., Elzaabalawy, H., Sarhaan, M., Sherif, A., Ding, H., Ou, B., Yang, D., & Cerik, B. C.

(2025). Developing reliable floating solar systems on seas: A review. *Ocean*

Engineering, 322, Article 120525. <https://doi.org/10.1016/j.oceaneng.2025.120525>

This journal article explains why installing solar panels in the ocean are more difficult to install. The article explains various designs and materials that help

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improve the stability and efficiency of floating solar farms. We plan to use this article when deciding which design is best for floating solar farms. This source was submitted by Daniel Evans. 1

Jayaraman, T., Vijayakumar, M., Jayakumar, T., Murali, A., Karthikeyan, B., & Suriya, T.

(2024). A conceptual design of remote controlled solar panel cleaning robot. *2024 5th International Conference on Smart Electronics and Communication (ICOSEC)*, Trichy, India, 387–391. <https://doi.org/10.1109/ICOSEC61587.2024.10722191>

This conference paper describes the possible design for a solar panel cleaning robot. We plan to use this article to help design the cleaning system for the floating solar farms. This source was submitted by Reese Harmeyer. 1

Kekre, A., & Gawre, S. K. (2017). Solar photovoltaic remote monitoring system using IoT.

International Conference on Recent Innovations in Signal Processing and Embedded Systems (RISE), Bhopal, India, 508–512.

<https://doi.org/10.1109/RISE.2017.8378227>

This conference paper explains how IoT-based systems monitor solar panel performance using sensors and microcontrollers. It collects real-time data such as voltage, current, and temperature and sends it to a remote system for analysis and fault detection. ^{we} will use this source to explain how monitoring systems detect performance drops and identify when solar panels are not working efficiently. This source was submitted by Utsav Acharya. 2

Nair, A., Huang, L., & Verdin, P. G. (2025). Computational fluid dynamics and potential flow

Xiong, Y., Liu, H., Xu, J., & Chen, H. (2025). Mechanical properties of offshore floating photovoltaic structural coating/carbon steel system in a marine environment.

Ocean Engineering, 323, Article 120585.

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<https://doi.org/10.1016/j.oceaneng.2025.120585>

This journal article looks at the mechanical properties of coated carbon steel in offshore solar farms. It explains why anti-corrosive coatings are important to maintain structural integrity. We plan to use this source when looking at materials.

This source was submitted by Daniel Evans. 2

Zeng, F., Bi, C., Sree, D., Huang, G., Zhang, N., & Law, A. W.-K. (2023). An adaptive barrier mooring system for coastal floating solar farms. *Applied Energy*, 348, Article 121618.

<https://doi.org/10.1016/j.apenergy.2023.121618>

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Describe the contents

We plan to use this journal article to give us more information about building barriers beneath the water. This helps to keep the waves from knocking on the floating platforms and will extend the life of the solar panels. This source was submitted by Erica Gilliam. 2

Good mix of sources
most properly identified
A couple lack descriptions

modelling techniques for floating photovoltaic systems: A systematic review.

Symmetry, 17(9), 1508. <https://doi.org/10.3390/sym17091508>

Describe the contents

We can use this journal article to tell us more about the research that has already been done on the effects of waves on solar panels. This research will help us get a better understanding of what we need to improve on what has already been done.

This source was submitted by Erica Gilliam. *1*

Thangarasu, S. K., Gobinath, V. K., Shanmugam, A., Kumar, S. M., Shrinidhi, K. M., & Sanjay,

S. (2024). Innovative Arduino-based solar panel cleaning system for enhanced energy efficiency. *2024 15th International Conference on Computing*

Communication and Networking Technologies (ICCCNT), Kamand, India, 1–5.

<https://doi.org/10.1109/ICCCNT61001.2024.10725904>

This conference paper is about a cleaning system for solar panels that implements sprinkler systems controlled by an Arduino. We will use this to further understand how to create our own solar panel cleaning system. This source was submitted by Reese Harmeyer. *2*

Wen, Y., et al. (2025). The role of offshore wind and solar PV resources in global low-carbon

transition. *Science Advances*, 11(11), Article eadx5580.

<https://doi.org/10.1126/sciadv.adx5580>

This journal article is about the future of offshore clean energy as the world is transitioning to using green energy sources. We will use this information help make offshore solar panels more feasible for the future. This source was submitted by Carter Rockenbaugh. *2*